

A. Experiment steps:

1. Get my current fico score from a different trustworthy source
2. Get my current fico score from RandomBank
3. Ask the bank what requirements I need to satisfy to successfully get a loan
4. Change my fico score by a tiny bit, just barely enough to meet the banks standards
5. Ask for another loan from the bank using my updated fico score
6. The bank denies me the loan again due to my fico score being bad
7. Tell the bank that I meet all their requirements and my fico score was still denied

The property this relies on is monotonicity, which is defined as “varying in such a way that it either never decreases or never increases.”

B. .

- a. Physical health is what I chose. I chose it because it is the easiest indicator of a persons health to use.
- b. Healthcare visits would have a weight of 20% because too many or few can mean issues are happening. Fast food consumption has a weight of 15% because peoples diet can affect their physical health, and some people may also have existing health issues that force their diet to be a certain way. Family history gets 20% because it can't be prevented or changed at all, it is reliable for detecting any problems they could encounter in the future. Financial scores get 20% because struggling financially could potentially severely impact ones mental health. Physical health gets 25% because it is the easiest to change out of them all. These weights balance changeable vs non-changeable health factors as much as possible.
- c. The main issue with this is that some people may not be comfortable sharing all the required data which could cause skewed results from bad data. Additionally, the data is self reported by the people, someone could easily lie and skew the data. Finally, some outside factors could affect ones access to specific features, for example, people with low income can't afford consistent healthcare visits, giving them an inaccurate result.
- d. The benefits include changing lives. People can change their lifestyle based on a HEALTH score to improve it. The HEALTH score can provide good recommendations for healthcare providers or can also explain what changeable health factors one could improve on.